

Agenda

Modules

OOPs

Ai assisted coding – vibe coding & vibe engg

Misc – Iterators, Generators, lambda Exp,
thirdparty modules

Modules and Packages

Lab: Create a module & a package for ip, email utilities

Functions

builtin Modules

Batteries included

re

os

sys

csv

json

urllib3

sockets

string

file handling

Scenario: You have got a folder with different files. You want to segregate them into folders based on extension

os

pathlib

```
=====
Character Meaning
-----
'r'      open for reading (default)
'w'      open for writing, truncating the file first
'x'      create a new file and open it for writing
'a'      open for writing, appending to the end of the file if it exists
'b'      binary mode
't'      text mode (default)
'+'      open a disk file for updating (reading and writing)
=====
```

Object Oriented Programming

Which are the 4 pillars of OOPs?

Encapsulation

→ Attributes and their methods together in a class

Abstraction

Inheritance

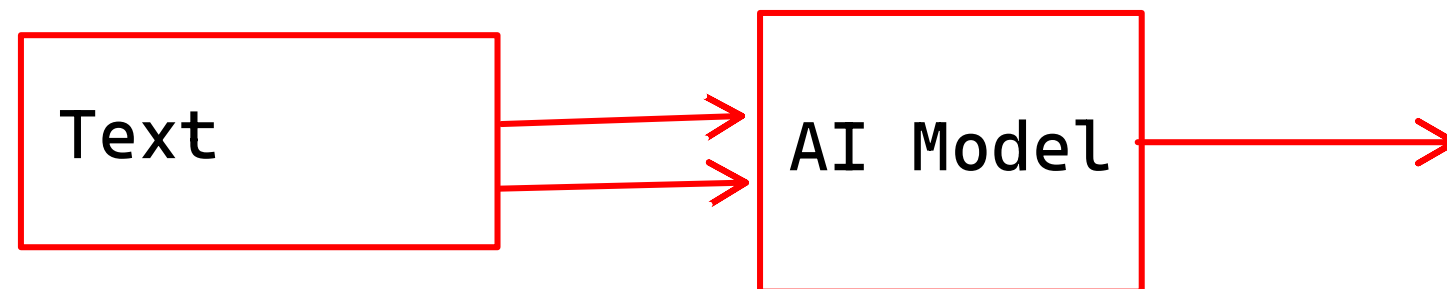
Polymorphism

Which is a special method which is run when an object is instantiated in OOPs?

AI Assisted Development

Claude, ChatGPT, Gemini.... → Transformers

Interact with Natural Language (NLP)

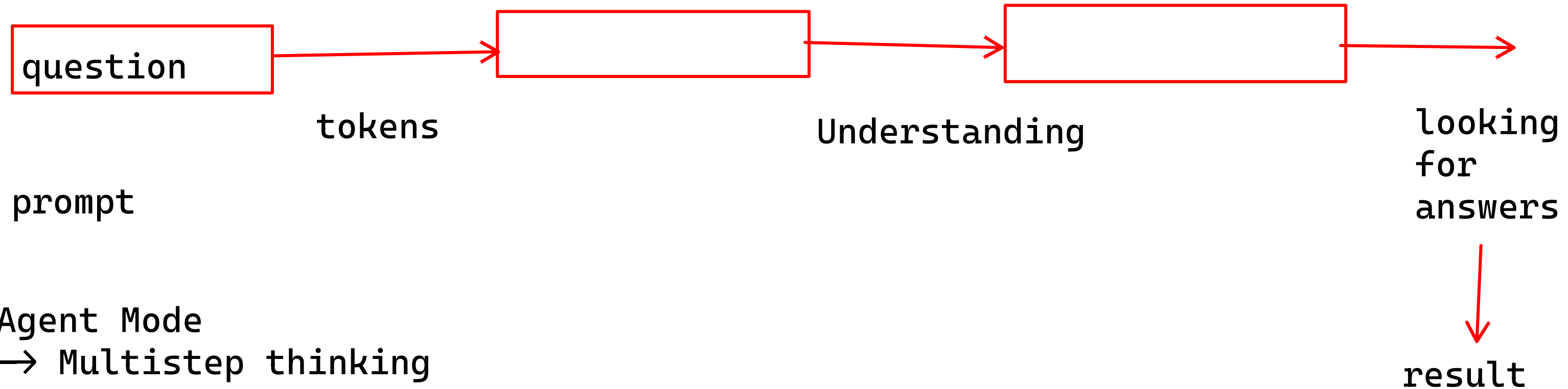


Sequence

This is cat → [0.1, 1.2, 2.6]

Tokens

What is cloud Computing?



Agent Mode

- Multistep thinking
- Multi Turn Thinking
- Make decisions
- Retry loops
- has access to few tools builtin

The way in which we give instructions → Prompt

- Few-shot Prompting
- Role Prompting
- System/Context Prompting
- Chain of Thoughts

Few-shot prompting → While Prompting give few examples

Classify the sentiment of customer feedback into Positive, Negative, or Neutral

Example 1:

Feedback: "The delivery took 5 days instead of 2, and the box was dented"

Sentiment: Negative

Example 2:

Feedback: "Received the item today, works as expected"

Sentiment: Neutral

Role Prompting

Act as a Principal Senior Security Architect performing a code review
Your goal is to identify security risks in the provided snippet

Constraints:

....

Code Snippet:

...

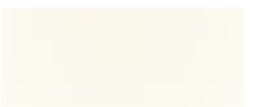
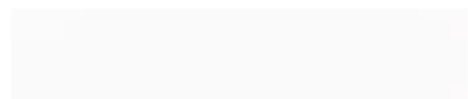
System Prompting

SYSTEM CONTEXT:

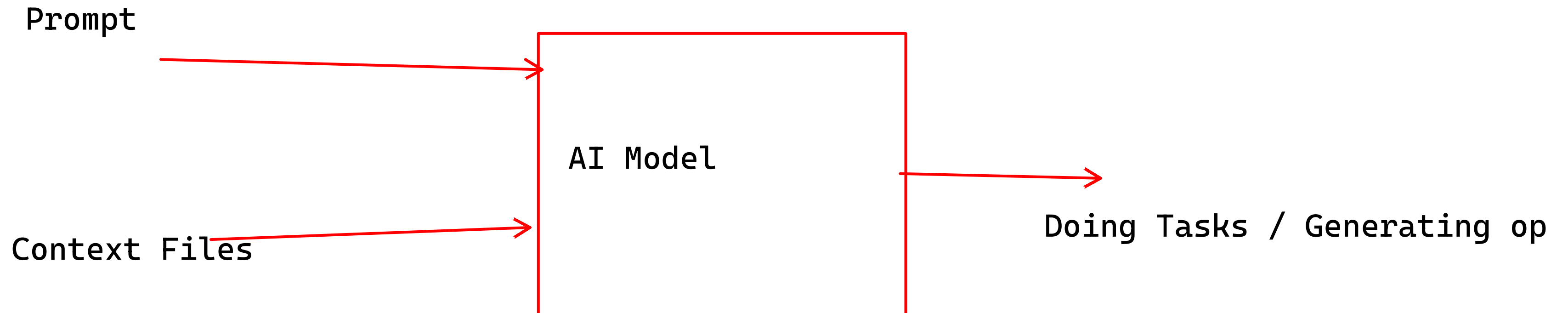
You are a support agent for A[]. Answer the user's questions STRICTLY using only the provided documentation below. If the answer cannot be derived directly respond with "I DO NOT HAVE ENOUGH INFORMATION IN MY base to ANSWER the question.

DOCUMENTATION:

.....



Vibe coding → Vibe Engineering



OOPs in Python

- Encapsulation
- inheritance
- Polymorphism (Runtime)

Access modifiers

Python does not provide direct boundaries

- Private/protected variable/method inside a class → through conventions

`_variable` → Protected

`__variable` → Private

Decorators, iterators, generators & lambda Functions

lambda Functions

→ Short, anonymous functions

```
lambda a,v: a**v
```

```
>>> inventory
{'mouse': 10, 'keyboard': 200, 'laptop': 5, 'monitor': 2}
>>> inv3 = list(inventory.items())
>>> inv3
[('mouse', 10), ('keyboard', 200), ('laptop', 5), ('monitor', 2)]
>>> inv3.sort(key= lambda x: x[1])
>>> inv3
[('monitor', 2), ('laptop', 5), ('mouse', 10), ('keyboard', 200)]
>>>
```

```
>>> sq = lambda x: x**2
>>> sq(6)
36
>>> sq(1000)
1000000
>>> inventory = {"Mouse":100, "keyboard":2, "laptop":10, "monitor":50}
>>> # print in terms of ascending order of number of items
>>> inv1 = inventory.copy()
>>> for i in inv1.keys():
...     print(i)
...
Mouse
keyboard
laptop
monitor
>>> inv_keys = inv1.sort()
Traceback (most recent call last):
  File "<python-input-27>", line 1, in <module>
    inv_keys = inv1.sort()
                ^^^^^^^^^
AttributeError: 'dict' object has no attribute 'sort'
>>> inv_keys = list(inv1.keys())
>>> inv_keys
['Mouse', 'keyboard', 'laptop', 'monitor']
>>> inv_keys.sort()
>>> inv_keys
['Mouse', 'keyboard', 'laptop', 'monitor']
>>> |
```

iterators

iterator → Object which helps in iterating over iterable

iterable → sequence → list, tuple, dict, set....

```
>>> l1
['23', '45', '96', '2']
>>> it = iter(l1)
>>> next(it)
'23'
>>> next(it)
'45'
>>> next(it)
'96'
>>> next(it)
'2'
>>> next(it)
Traceback (most recent call last):
  File "<python-input-58>", line 1, in <module>
    next(it)
    ~~~~^
StopIteration
>>>
```

Generator

Function which gives us value when asked

```
>>> def numgen(n):  
...     for i in range(n):  
...         yield i  
...  
>>> k = numgen(10_000_000)  
>>> next(k)  
0  
>>> next(k)  
1  
>>> next(k)  
2  
>>> next(k)  
3  
>>> next(k)  
4  
>>> next(k)  
5  
>>> next(k)  
6  
>>> next(k)  
7  
>>> next(k)  
8  
>>> next(k)  
9  
>>> next(k)  
10  
>>> next(k)  
11  
>>> next(k)  
12  
>>> next(k)  
13  
>>> |
```